AI-FIRST DELIVERY PARTNERSHIP

Trinly × ArtSync

Work Portfolio — AI systems we have designed, built and shipped

Five production platforms across clinical trials, drone inspection, railway safety, enterprise integrity and commercial web — plus the full service and support capability behind them.



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01

Who we are

One allied delivery group covering the complete path from raw data to a production AI product — senior strategy from the US and Europe, deep engineering and data capacity from India.

Trinly

US · AI CONSULTANCY

AI-first technology consultancy turning AI ambitions into shipped, production-ready products — RAG systems, model training, AI agents and MLOps. Architecture, strategy and delivery leadership.

trinly.us

ArtSync India

INDIA · AI & DATA INTELLIGENCE

AI development, data intelligence and annotation at scale — data engineering, ML/LLM engineering, automation, enterprise applications and ongoing IT support from six hubs across India.

artsync.live

17+
AI engineers & architects

6
Global delivery hubs

5
Countries

24/7
Follow-the-sun coverage

02

What we do

Eight capabilities, each one demonstrated by a project later in this deck.

RAG & retrieval systems

Production retrieval pipelines with vector search, semantic precedent lookup and context-aware grounding.

Model training & fine-tuning

Custom training, LoRA and domain adaptation — with evaluation harnesses and retraining loops.

AI agents & orchestration

Multi-agent, tool-using systems that carry out multi-step business workflows end to end.

Computer vision

Detection and segmentation on visual and thermal imagery, at the edge and in the cloud.

Data engineering & annotation

ETL, cleaning, labeling and ML-ready dataset delivery — including physical-AI video annotation.

MLOps & deployment

Serving, monitoring, drift detection, A/B testing and continuous deployment.

Full-stack product engineering

Web, native mobile and API platforms — monorepos, design systems, CI/CD.

Regulated & compliant systems

Audit trails, e-signatures, RBAC and submission packages for regulated industries.

03

Selected work

Five systems, deliberately spanning the range — from a regulated clinical data platform to a contractor website shipped in weeks.

PROJECT	DOMAIN	WHAT IT IS
Helix	Clinical trials · regulated life sciences	Decision-intelligence platform for the ODM → SDTM → ADaM pipeline, with AI-assisted mapping and 21 CFR Part 11 compliance.
Sentinel	Drone · infrastructure inspection	Drones survey the asset, computer vision finds the defects, the platform turns them into tracked, SLA-bound work orders.
Cortex	Railway · edge AI	Cart-mounted Raspberry Pi captures track imagery over 4G; cloud AI detects rail defects on a geospatial operator dashboard.
RDACTD	Enterprise integrity · SaaS	Encrypted complaint management, resolution workflows and organizational integrity analytics — web and native mobile.
Open Air Construction	Commercial web · local SEO	Contractor marketing site with typed content, MDX blog, lead pipeline and an admin inbox — Lighthouse SEO 100.

04

Helix

Institutional memory for clinical data mapping

CLINICAL TRIALS · REGULATED LIFE SCIENCES



THE CONTEXT

Decision Intelligence Platform for Clinical Data — capturing, versioning and reusing mapping decisions across the ODM → SDTM → ADaM transformation pipeline.

THE PROBLEM

Every clinical study re-derives the same data-mapping decisions from scratch. The reasoning lives in spreadsheets and in people's heads, it is lost when staff move on, and regulators still expect a defensible audit trail for every one of them.

WHAT WE BUILT

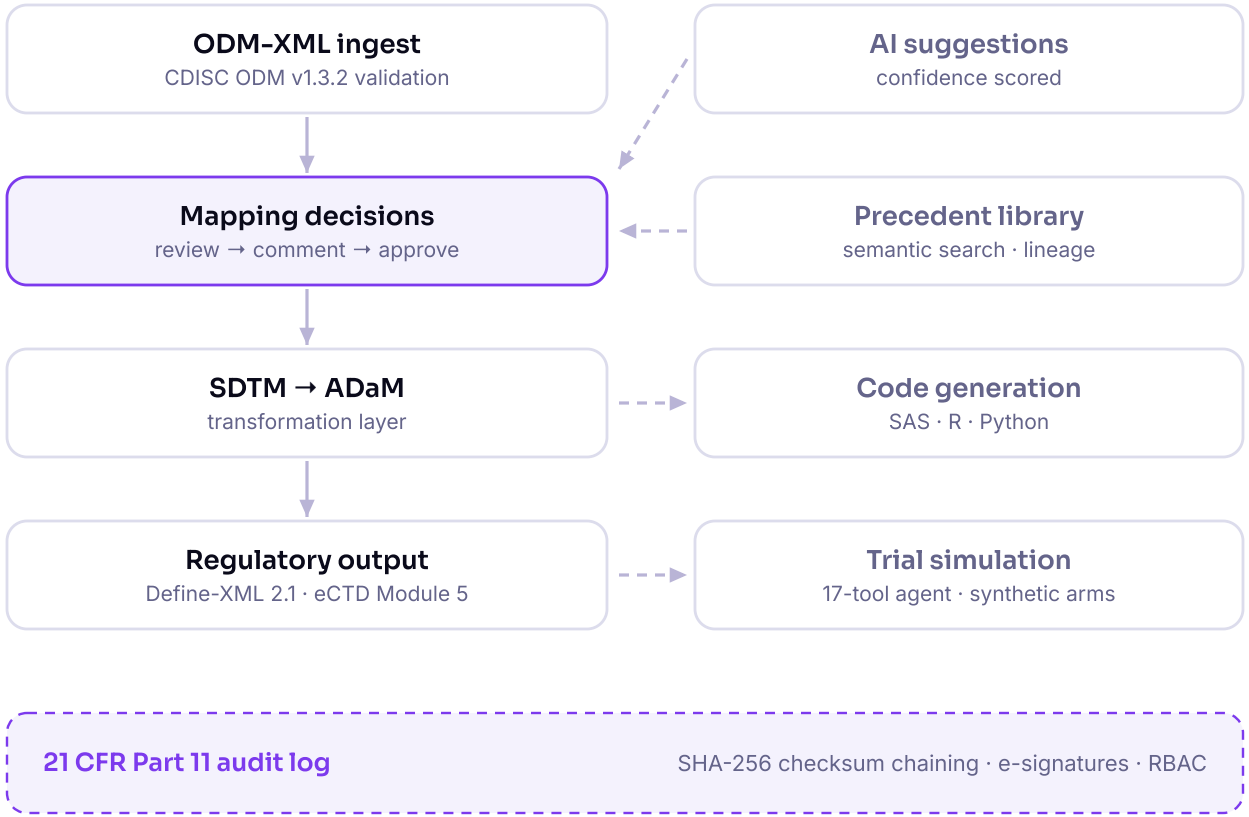
- CDISC ODM v1.3.2 ingestion with schema validation, then mapping decisions carried through review, comment and approval chains
- Cross-study precedent reuse — semantic search over prior decisions, copied forward with lineage tracking and conflict resolution
- AI-assisted mapping suggestions with confidence scoring, plus nine anomaly detectors for cross-study inconsistency
- Executable SAS, R and Python code generated straight from approved decisions
- Virtual trial simulation: a 17-tool agent runs synthetic cohorts visit by visit, with synthetic control arms and ML enrollment forecasting
- Regulatory output — eCTD Module 5 packages, Define-XML 2.1, ClinicalTrials.gov registration, RBQM risk dashboards

WHY IT IS HARD

21 CFR Part 11 compliance is not a feature you bolt on. Every action writes to an append-only audit log chained with SHA-256 checksums, backed by electronic signatures and role-based access control — so the record is tamper-evident end to end.

Helix

Architecture & stack



STACK

BACKEND	FastAPI · Python 3.12 · SQLAlchemy · Alembic
DATA	PostgreSQL · vector embeddings for precedent search
FRONTEND	Expo Web (React Native Web) · shared RN component library
MOBILE	Expo native app · biometric auth · mobile approvals
AI	Multi-provider LLM layer — OpenAI, Azure, Anthropic, Ollama
MONOREPO	pnpm workspaces · apps/api, apps/web, apps/mobile, packages/*

NOTABLE

- → Compliance by construction — append-only audit log, SHA-256 checksum chaining, e-signatures, RBAC
- → Integration hub with bidirectional connectors for Medidata Rave, Veeva Vault, Oracle InForm and SAS
- → Real-world data connectors — FDA FAERS, openFDA, PubMed, CMS — with OMOP CDM and FHIR R4 support
- → Eight interface languages with multilingual embeddings, so precedent search works across regions

05

Sentinel

From drone survey to closed work order

DRONE OPERATIONS · INFRASTRUCTURE INSPECTION



THE CONTEXT

AI-powered infrastructure inspection platform. Drones fly the asset, computer vision finds the defects, and the platform turns them into tracked, assigned, SLA-bound work.

THE PROBLEM

Infrastructure inspection is manual, slow and hazardous. The findings end up in photo folders and spreadsheets, disconnected from the crews who have to fix anything — so nothing is provably closed out.

WHAT WE BUILT

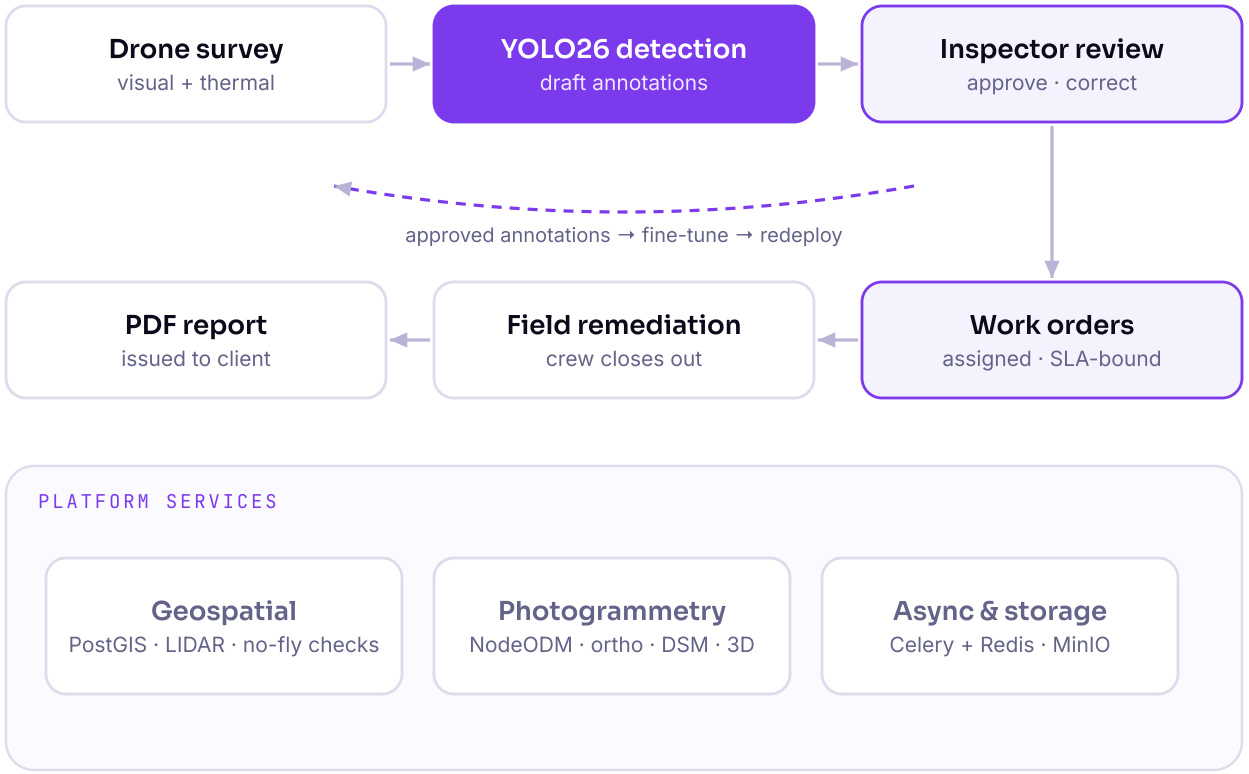
- The core loop: survey → visual and thermal imagery → YOLO26 detection → draft annotations → inspector review → work orders → field remediation → PDF report
- Geospatial context — KML/Shapefile/GeoJSON import, LIDAR point clouds, corridor encroachment and terrain-risk analytics
- Photogrammetry via NodeODM: orthophotos, digital surface models and textured 3D models, served as map tiles and an in-browser point-cloud viewer
- Drone operations — flight planning with safety validation, no-fly checks against FAA/AirMap, autonomous and swarm missions
- A PX4 SITL simulator so missions are tested in software before any aircraft leaves the ground
- Web, native mobile and a public marketing site, all from one Nx monorepo

WHY IT IS HARD

The AI improves itself. Every annotation an inspector approves becomes labelled training data; the model is fine-tuned and redeployed. Human review is not a cost centre in this system — it is the training pipeline.

Sentinel

Architecture & stack



STACK

FRONTEND	React 19 · TypeScript 5.5 · Vite 6 · Tailwind 4 · Leaflet · Three.js · Konva
BACKEND	Python 3.11 · FastAPI · SQLAlchemy 2 · Pydantic 2 · Alembic
DATABASE	PostgreSQL + PostGIS
ASYNC	Celery + Redis — queues for AI, reports, work orders, LIDAR
AI & IMAGING	YOLO26 via Ultralytics (separate visual and thermal models) · NodeODM
PLATFORM	MinIO object storage · React Native/Expo mobile · Astro marketing · Nx

NOTABLE

- → Closed learning loop — approved annotations retrain the detector and redeploy it
- → Separate visual and thermal models, because a hot joint and a cracked insulator are not the same detection problem
- → Published documentation site built from the repository and deployed on every merge
- → Simulation-first drone testing with PX4 SITL — no hardware required to validate a mission

06

Cortex

AI track inspection from a cart and a Raspberry Pi

RAILWAY SAFETY · EDGE AI



THE CONTEXT

Railway track inspection system. A cart-mounted Raspberry Pi captures imagery during inspection runs, uploads it over 4G, and cloud AI identifies track defects in near real time.

THE PROBLEM

Track is still inspected largely on foot. Coverage is slow, defects are found late, and there is no consistent geolocated record of what was seen on which run — which makes trends across a corridor invisible.

WHAT WE BUILT

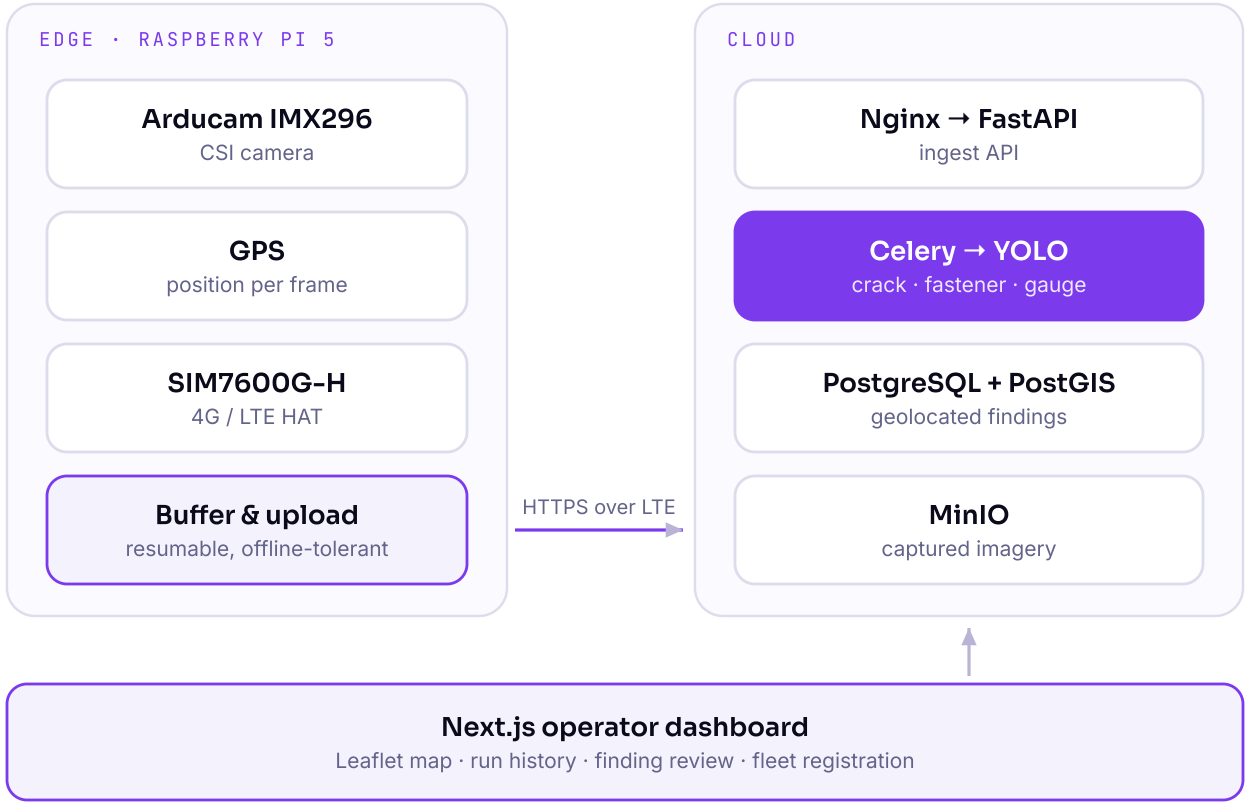
- An edge capture daemon on a Raspberry Pi 5 with an Arducam IMX296 camera, GPS and a 4G HAT — geotagged imagery captured on the move
- Local buffering and resumable upload over LTE, so a run in a dead zone is never a lost run
- Cloud detection pipeline — FastAPI and Celery workers running YOLO models for rail cracks, missing fasteners and gauge deviations
- PostGIS-backed findings, so every defect carries its position on the track segment
- A Next.js operator dashboard with a Leaflet map, run history and per-finding review
- Bot registration and API-key provisioning, so a new cart is added to the fleet in minutes

WHY IT IS HARD

Field connectivity is unreliable by default. The uploader's offline-resilience behaviour is covered by a test suite that runs on any laptop — no Pi, no camera, no track — so the hardest failure mode is verified continuously, not on site.

Cortex

Architecture & stack



STACK

EDGE	Raspberry Pi 5 · Python · picamera2 · pyserial · systemd daemon
HARDWARE	Arducam IMX296 (CSI) · GPS · SIM7600G-H 4G HAT
BACKEND	FastAPI · SQLAlchemy · Celery · YOLO detection workers
DATABASE	PostgreSQL + PostGIS for geolocated findings
STORAGE	MinIO (S3-compatible) for captured imagery
FRONTEND & OPS	Next.js 15 · TypeScript · Tailwind · Leaflet · Docker Compose · Nginx

NOTABLE

- → Runs on commodity hardware — a Pi, a camera and a 4G modem, not a specialised inspection vehicle
- → Offline-first by design: buffer on the cart, reconcile in the cloud
- → Single-command local environment (docker-compose up) with an auto-seeded admin user
- → Provisioning is scripted end to end, from flashing the Pi to starting the capture service

07

RDACTD

Complaints that can be trusted, tracked and measured

ENTERPRISE INTEGRITY · MULTI-TENANT SAAS



THE CONTEXT

Complaint management and business analytics platform — encrypted reporting, structured resolution, and integrity analytics for multi-branch organisations.

THE PROBLEM

Serious complaints still travel by email. Reporters have no privacy and no visibility, organisations have no structured record, and neither side can show afterwards that anything was actually resolved.

WHAT WE BUILT

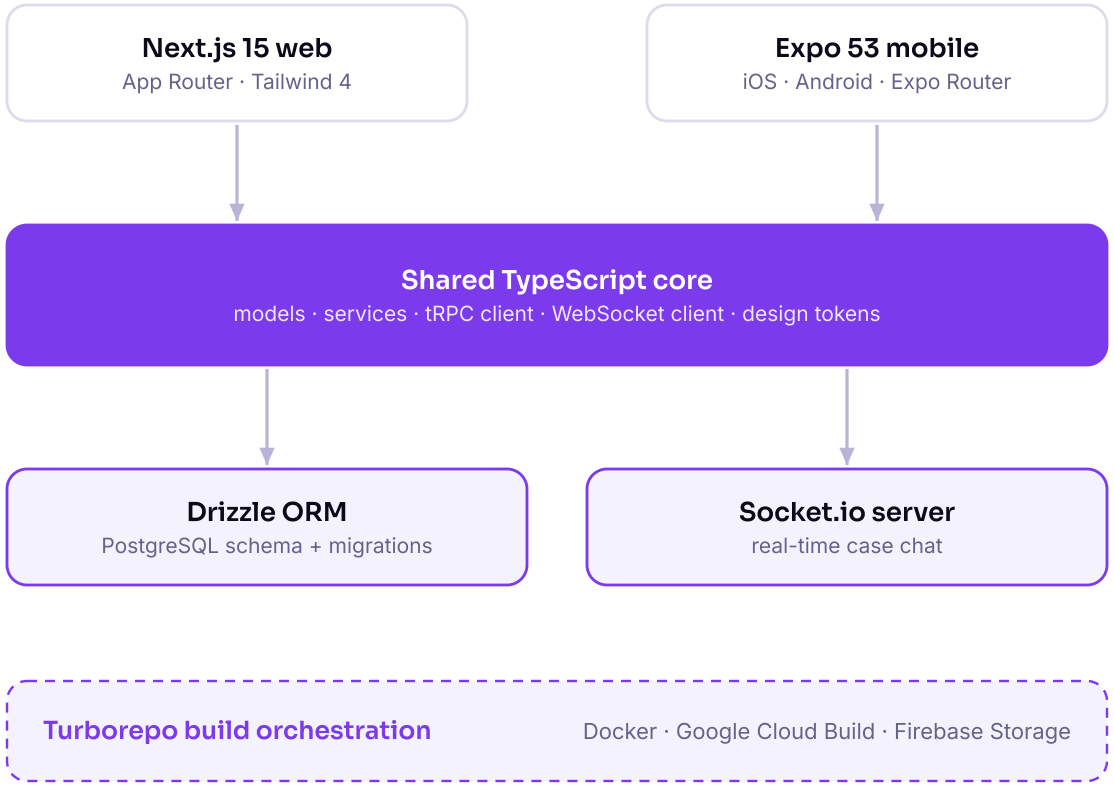
- Encrypted, evidence-backed submission flows with categorical routing and real-time status tracking
- End-to-end resolution channels that let a reporter and an organisation lead work a case to closure
- Organizational health dashboards for multi-branch entities, with predictive risk modelling and integrity benchmarking
- AI-assisted pattern identification across cases, plus automated structural-health forecasting
- Public verified profiles showing integrity scores, historical performance and resolution velocity
- Multi-tier SaaS framework with billing, team governance and role-based administration

WHY IT IS HARD

Web and native mobile share one TypeScript core — models, service layer, tRPC client, WebSocket client, Drizzle schema and design tokens all live in shared packages. The two clients cannot drift apart, because there is only one implementation of the rules.

RDACTD

Architecture & stack



STACK

WEB	Next.js 15 · Tailwind CSS 4 · SEO-ready App Router
MOBILE	Expo 53 · Expo Router · native iOS and Android
SHARED CORE	packages/shared — models, services, tRPC client, WebSocket client
DATA	packages/database — Drizzle ORM schema on PostgreSQL
REALTIME	packages/websocket — Socket.io server for live case chat
BUILD & DEPLOY	Turborepo · Docker · Google Cloud Build · Firebase Storage

NOTABLE

- → One monorepo, four packages, two apps — a change to a business rule lands everywhere at once
- → Design tokens as a package, so web and native render the same product
- → Real-time case chat over Socket.io, with offline-tolerant client behaviour
- → Containerised web and websocket services, built and deployed through Cloud Build pipelines

08

Open Air Construction

A contractor site built to convert, not just to exist

COMMERCIAL WEB · LEAD GENERATION · LOCAL SEO



THE CONTEXT

Charlotte deck and outdoor-living contractor website — a production marketing site with a full lead pipeline behind it.

THE PROBLEM

A local contractor needs more than a brochure: fast pages, trust signals, service-area coverage, and a dependable way to turn consultation requests into leads someone actually follows up — without losing editorial control or SEO.

WHAT WE BUILT

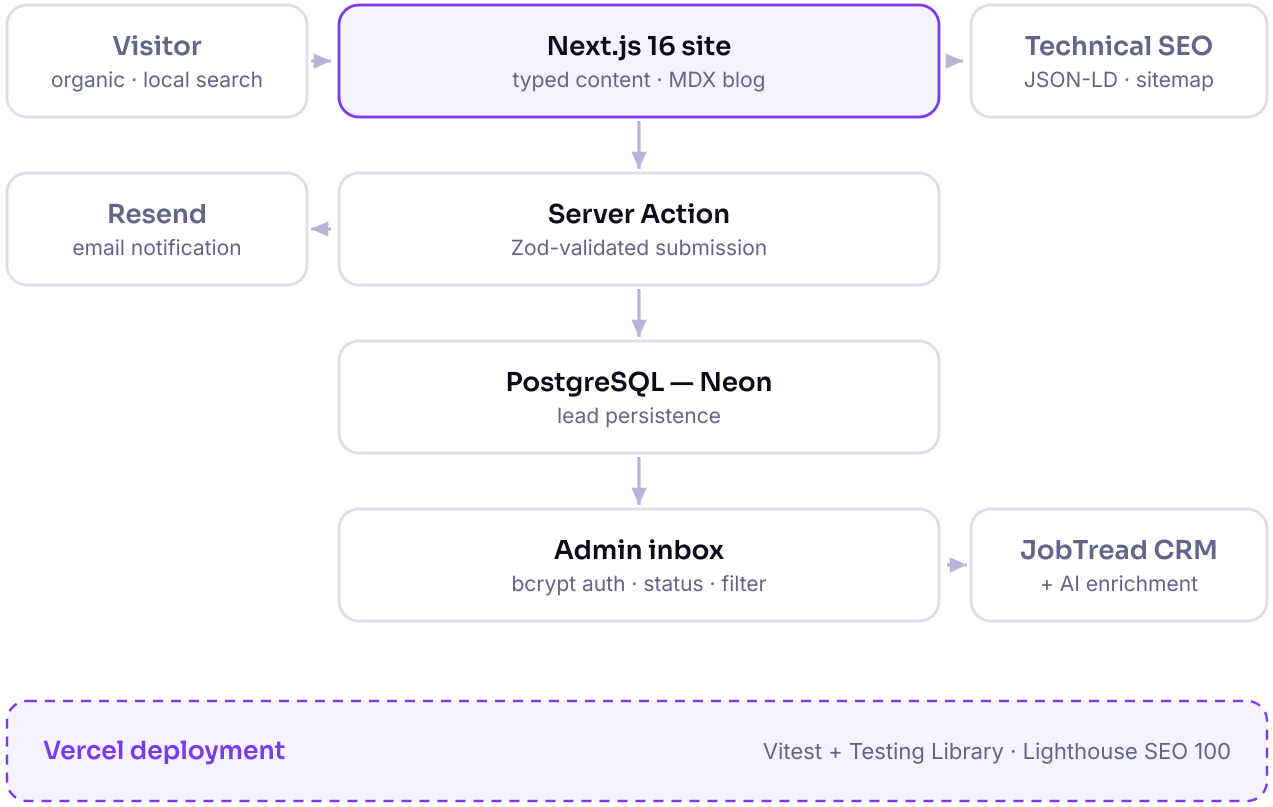
- Next.js 16 App Router with Tailwind CSS v4 over a custom Material 3 token layer, light and dark themes
- A typed content layer — services, team, FAQs and metadata in TypeScript, so structural mistakes fail the build
- An MDX blog the owner can add to by dropping in a file, rendered through a custom loader
- Lead pipeline: contact form Server Action with Zod validation, Postgres persistence and Resend email notification
- A protected admin inbox with bcrypt auth, lead filtering, status management and CRM sync hooks
- Technical SEO — LocalBusiness, Service and FAQ JSON-LD, sitemap, robots, dynamic OG images

THE RESULT

A Lighthouse SEO score of 100, an editor workflow that needs no developer, and JobTread CRM plus AI enrichment already wired in behind environment keys — they switch on without touching the plumbing.

Open Air Construction

Architecture & stack



STACK

FRONTEND	Next.js 16 · TypeScript · Tailwind CSS v4 · Material 3 tokens · next-themes
CONTENT	Typed TS content modules · MDX blog via next-mdx-remote/rsc
BACKEND	Server Actions · Zod validation · PostgreSQL (Neon)
EMAIL & CRM	Resend notifications · JobTread sync seam · AI enrichment gateway
SEO	JSON-LD (LocalBusiness, Service, FAQ) · sitemap · robots · dynamic OG images
QUALITY & OPS	Vitest + Testing Library · Vercel deployment

NOTABLE

- → Lighthouse SEO score of 100 at launch
- → Non-developers publish blog posts and update service copy without touching layout code
- → Integrations activate by setting environment keys — no code change required
- → Proof that the same team that ships regulated AI platforms also ships a marketing site in weeks

09

Trinly — services

AI strategy through to shipped product. Delivered from the US and Europe, with the India engineering bench behind every engagement.

AI strategy & consulting

Where AI actually pays in your business, what to build first, and what to leave alone.

RAG & retrieval systems

Chunking, embeddings, vector search, reranking and evaluation — grounded answers with citations.

Model training & fine-tuning

Custom training, LoRA, domain adaptation, and the eval harness that proves it improved.

AI agents & automation

Tool-using agents that complete multi-step workflows, with guardrails and human checkpoints.

Computer vision

Detection, segmentation and thermal analysis — cloud pipelines or constrained edge devices.

Data pipelines & platform

Ingestion, ETL, geospatial and time-series stores, and the infrastructure underneath.

MLOps & production

Serving, monitoring, drift detection, retraining loops, CI/CD and cost control.

Web & mobile product

Full product engineering — Next.js and React web, React Native and Expo mobile, APIs and admin tooling.

10

ArtSync India — services & support

Your partner in digital excellence: software development, data intelligence and dependable IT support, delivered at scale from six hubs across India.

Custom software development

Web applications, enterprise systems and internal tools, built and maintained by dedicated teams.

Enterprise solutions & integration

ERP, CRM and line-of-business integration; APIs and middleware between systems that were never designed to talk.

Data intelligence & engineering

Data cleaning, transformation, migration and warehousing — turning messy source data into something models can use.

Data annotation at scale

Image, video, text and multimodal labeling with human-in-the-loop QA and ontology governance.

Data entry & back-office operations

High-volume, accuracy-critical processing with defined SLAs and quality sampling.

IT support & managed services

Helpdesk, infrastructure monitoring, patching and maintenance — day-to-day operational cover.

11

Specialisation — data annotation for physical AI

A core group specialisation: high-quality video and multimodal annotation for computer-vision and Vision-Language-Action systems.

Robotics and physical-AI teams are bottlenecked on demonstration data, not on model architecture. We deliver frame-accurate temporal segmentation, keypoint and object-interaction labeling, multi-view clip decomposition, and ontology-driven consistency across large demonstration datasets — backed by in-house ML teams, so datasets arrive clean, structured and training-ready rather than as a pile of raw labels someone else has to fix.

Frame-accurate temporal segmentation
Action boundaries labelled to the frame, not the second.

Multi-view decomposition
Synchronised labeling across camera rigs for the same demonstration.

Human-in-the-loop QA
Sampled review, inter-annotator agreement and correction loops.

Keypoint & object interaction
Grasp points, contact events and manipulation states across the clip.

Ontology governance
One label taxonomy, enforced across every annotator and every batch.

ML teams in the loop
The people labeling the data work alongside the people training on it.

CVAT · Labelbox · Label Studio · Temporal segmentation · Keypoint labeling · HITL QA · VLA datasets

12

How we deliver

The same four-stage path on every engagement — from a two-week website to a regulated clinical platform. Only the depth changes.

01

Data audit & strategy

Audit what you have, structure it, and define an approach tied to a business outcome — not to a model.

02

Build & train

Model selection, training, pipeline and application build, with evaluation gates before anything is called done.

03

Production deployment

Scalable, monitored delivery to production — infrastructure, CI/CD, security and handover documentation.

04

Continuous optimisation

Live telemetry drives retraining, tuning and the next increment of the roadmap.

13

Engagement & ongoing support

Four ways to work with us, and a support commitment that does not end at launch.

ENGAGEMENT MODELS

Fixed-scope project

Defined deliverable, defined price, defined timeline. Best when the outcome is clear.

Retained / dedicated team

A named team on a monthly retainer, working your roadmap as priorities shift.

Staff augmentation

Senior AI, data and full-stack engineers embedded in your existing team.

Managed support & SLA

We operate what we built — or what someone else built — under an agreed response SLA.

WHAT SUPPORT INCLUDES

Production monitoring

Uptime, error and performance monitoring with defined incident response.

Model health

Drift detection, evaluation on live data, and a scheduled retraining cadence.

Security & maintenance

Dependency patching, vulnerability response, access reviews and backups.

Cost optimisation

Inference, storage and cloud spend reviewed and tuned as usage grows.

Enablement

Documentation, handover and training so your team can run the system themselves.

14

Industries & why clients choose us

INDUSTRIES WE SERVE

- Healthcare & clinical trials
- Rail & transport infrastructure
- Drone & geospatial
- Robotics & physical AI
- Financial services & fintech
- B2B SaaS & enterprise software
- Construction & home services

From early-stage startups to established enterprises.

WHY CLIENTS CHOOSE US

End-to-end ownership

Raw data to deployed product, with no hand-off gap between strategy, data and engineering.

Regulated-domain experience

We have built to 21 CFR Part 11 with tamper-evident audit trails and electronic signatures.

Senior plus scale

US and Europe based architects, with a deep India engineering and data bench behind them.

Production-first

Monitored systems running in production — not proofs of concept that stall at the demo.

Follow-the-sun delivery

Hubs across five countries keep work moving while your team sleeps.

LET'S TALK

Tell us what you are trying to build

Every engagement starts the same way: a 15-minute scoping call, then a written proposal with scope, timeline and price. No obligation, and no charge for the conversation.

Ashish Tyagi — Founder & CEO

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